Project Report

GROOMING AND RECOMMENDATION SYSTEM

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INTRODUCTION

Haircare is an important part of personal hygiene and grooming. Different people have different hair types such as oily, dry, normal, and combination. Identifying the correct hair type is essential to maintain healthy hair and avoid damage caused by unsuitable products.

This project is a Python-based application that helps users identify their hair type through a simple questionnaire and provides suitable grooming tips along with recommendations of chemicals to use and avoid.

2. Objective

- To identify hair type using user input
- To provide personalized haircare suggestions
- To recommend safe and harmful chemical ingredients
- To demonstrate the use of Python programming in real-life applications

3. Methodology

Step 1: Input Collection

The program asks the user a set of questions related to hair condition such as oillliness, scalp type, hair ends, and washing frequency.

Step 2: Scoring System

Each answer contributes to a score for different hair types (oily, dry, normal, combination).

Step 3: Hair Type Detection

The hair type with the highest score is selected using Python's max() function.

Step 4: Recommendation System

Based on the detected hair type, the program provides:

- Haircare tips
- Chemicals to avoid
- Recommended ingredients

4. Tools and Technologies Used

- **Programming Language:** Pythone
- **Concepts Used:**
    - Conditional statements (if-else)
    - Dictionaries
    - User input handling

5. Features

- Interactive command-line interface
- Quick and simpl hair type detection
- Personalized recommendations
- Easy to understand and modify

6. Working of the Project

1. The user runs the Python program
2. The system asks multiple-choice questions
3. The user selects answers
4. Scores are are calculated internally
5. Hair type is displayed
6. Haircare tips and chemical suggestions are shown

7. Advantages

- Simple and user-friendly
- No external libraries required
- Useful for beginners
- Practical real-life application

8. Limitations

- Not based on machine learning
- Uses fixed rules (no learning capability)
- Limited number of questions
- Accuracy depends on user responses

9. Future Scope

- Convert into an AI/ML-based system
- Add graphical user interface (GUI)
- Increase dataset and improve accuracy
- Add product recommendations
- Integrate mobile or web application

10. Conclusion

The Haircare Test and and Recommendation System is a simple and effective Python project that helps users identify their hair type and provides useful haircare advice. It demonstrates how programming can be used to solve everyday problems. Although the current system is rule-based, it can be further improved by integrating machine learning techniques for better accuracy and personalization.

11. References

- Python Documentation
- Basic Haircare Knowledge Sources
- Online tutorials on Python programming